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FINAL ASSESSMENT
II YEAR – SEMESTER 4
CSE23AE304 - PCP III
(B.Sc. E61)

SET G

Question 1: Problem Statement

An array of non-negative integers represents the **maximum jump length** that can be made from each index.

Starting at the **first index**, determine the **minimum number of jumps required to reach the last index**.

From position i , a jump can reach any position in the range:

$i + 1$ to $i + \text{nums}[i]$

It is guaranteed that the last index is reachable.

Input Format

`{"nums": [a1, a2, a3, ...]}`

Where:

- `nums[i]` represents the **maximum jump length** from index i .

Output Format

Return:

`minimum_jumps`

Where:

- `minimum_jumps` → minimum number of jumps needed to reach the last index.

Example 1

Input

`{"nums": [2,3,1,1,4]}`

Output

2

Explanation

Optimal path:

index 0 → jump to index 1

index 1 → jump to index 4

Minimum jumps = 2

Question 2: Problem Statement

A company has a network of **n servers**, numbered from **0 to $n-1$** .

The dependencies between servers are given as a list of directed edges:

`edges = [[u1, v1], [u2, v2], ...]`

Each edge `[u, v]` means:

- Server **u can activate server v**

Some servers depend on others, while some can operate independently.

Your task is to determine the **minimum set of servers that must be started manually** so that **all servers in the network become active**.

Return the list of such servers in **ascending order**.

Example 1

Input

`n = 6`

`edges = [[0,1],[0,2],[2,5]]`

Output

`[0,3,4]`

Explanation

`0 → 1`

`0 → 2 → 5`

3

4

Servers **3 and 4** are not reachable from any

Question 3: Problem Statement

In a distributed network router, packets are placed in a queue represented by an integer array:

- A **positive integer** → valid packet
- A **zero (0)** → empty slot (no packet received at that millisecond)

To optimize throughput, the system must **shift all valid packets to the front of the queue**, while keeping their **original order** unchanged. All empty slots (0s) must be moved to the **end**.

Routing Rule:

- **Do NOT create a new array**
- **Do NOT change relative order of valid packets**
- **Move zeros to the end in-place**

Example**Input:** [0, 5, 0, 3, 12]**Output:**

[5, 3, 12, 0, 0]

Explanation:

Process:

Valid packets: [5, 3, 12]

Zeros: [0, 0]

Question 4: Problem Statement

You are given a grid with m rows and n columns.

A robot starts at the **top-left corner (0,0)** and wants to reach the **bottom-right corner (m-1, n-1)**.

The robot can only move:

- Right →
- Down ↓

Your task is to find the **total number of unique paths** the robot can take to reach the destination.

Return the total number of unique paths.

Example 1

Input: m = 3, n = 7

Output: 28

Explanation:

Multiple paths exist by combining right and down moves

Question 5: Problem Statement

You are part of a **capacity analysis team** in an enterprise organization.

A legacy analytics module generates a **numeric utilization snapshot** that visually expands and contracts based on a given scale value.

Each row of the snapshot reflects **row-level indexing** and must be stored as text for automated verification.

Since the analytics engine validates output programmatically, the snapshot must be returned as a **list/array of strings**, where:

- Each string represents **one row**
- Numbers in a row correspond to the **row index**
- The structure is vertically symmetric
- Alignment must be preserved exactly

Your task is to generate this snapshot for a given integer N.

Example 1**Input**

3

Output

```
[  
  "33333",  
  " 222 ",  
  "   1   ",  
  " 222 ",  
  "33333"  
]
```